

SOC ANALYST PORTFOLIO CASE STUDY

Authorized Network and Service Enumeration

Nmap reconnaissance, SSH cryptographic assessment, source validation, and evidence integrity

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OUTCOME Scoped one authorized Ubuntu VM, confirmed host availability, identified SSH and Apache exposure, assessed protocol configuration, validated automated findings on the endpoint, and preserved six evidence files with SHA-256 hashes.

Component	Address / role	Control
Parrot Security 7.3	10.10.10.5 - assessment workstation	Nmap 7.95 and SSH-Audit
Ubuntu Server 24.04.4 LTS	10.10.10.4 - authorized target	UFW, OpenSSH, Apache
SOC-Lab	10.10.10.0/24 isolated NAT network	No public or production target

Portfolio note: Authorized local simulation only. No external or production system was targeted. AI assistance is disclosed in this report.

1. Executive Summary

I conducted a controlled network and service assessment from Parrot Security against a single Ubuntu web-server VM. Host discovery confirmed the target at 10.10.10.4. A SYN scan found only TCP/22 and TCP/80 open among the 100 most common TCP ports; the remaining 98 were filtered. Nmap identified OpenSSH 9.6p1 and Apache 2.4.58, while HTTP scripts exposed the default Apache page, precise server banner, and supported methods.

Nmap OS detection correctly inferred Linux but could not produce an exact match because firewall filtering prevented it from observing both an open and a closed TCP port. SSH-Audit found strong modern cryptography alongside legacy-compatible MAC and key-exchange options. I validated the scanner output directly on Ubuntu with `hostnamectl`, `ss`, `ssh`, `apache2`, and `sshd` effective-configuration commands. The assessment found no evidence of exploitation or critical compromise; it identified defensible hardening opportunities.

Assessment dimension	Observed result
Authorized scope	One owned VM: 10.10.10.4
Exposed services	TCP/22 OpenSSH; TCP/80 Apache
Firewall behavior	98 of top 100 TCP ports filtered
HTTP finding	Default page and precise Apache version disclosure
SSH finding	Modern algorithms available; unnecessary legacy MAC/KEX options remain
Disposition	Confirmed exposure and hardening opportunities; no compromise observed

2. Objectives

- Define and preserve a one-host authorized assessment scope.
- Perform host discovery, TCP port scanning, service identification, and OS fingerprinting.
- Enumerate HTTP behavior and assess SSH cryptographic configuration.
- Validate automated findings against authoritative endpoint evidence.
- Preserve scan output and generate cryptographic integrity hashes.

3. Scope and Safety Controls

Control	Implementation
Target authorization	Ryan Bryant's Ubuntu VM only
Target address	10.10.10.4/32
Scanner address	10.10.10.5/24
Network isolation	VirtualBox SOC-Lab NAT network
Scan timing	Nmap T3 normal controlled timing
Authentication behavior	No password guessing or login attempted
Evidence path	~/SOC-Labs/Nmap-Enumeration

4. Implementation Workflow

01 BASELINE

Confirmed the Parrot and Ubuntu addresses and required services.

02 HOST DISCOVERY

Used Nmap -sn against one IP and confirmed the target was up.

03 PORT ENUMERATION

Scanned the 100 most common TCP ports with a SYN scan.

04 SERVICE IDENTIFICATION

Used version detection and standard scripts on TCP/22 and TCP/80.

05 OS FINGERPRINTING

Evaluated Nmap's probabilistic Linux guesses and its stated test limitations.

06 HTTP ENUMERATION

Collected page title, headers, and advertised HTTP methods.

07 SSH ASSESSMENT

Used SSH-Audit to review KEX, host-key, cipher, and MAC algorithms.

08 SOURCE VALIDATION

Compared scanner results with Ubuntu's effective service and crypto configuration.

09 EVIDENCE INTEGRITY

Generated SHA-256 hashes for all six text evidence files.

5. Key Commands

```
sudo nmap -sn 10.10.10.4 -oN 01-host-discovery.txt
```

```
sudo nmap -sS -T3 --top-ports 100 --reason 10.10.10.4 -oN 02-top-ports.txt
```

```
sudo nmap -sV -sC -T3 -p 22,80 10.10.10.4 -oN 03-services.txt
```

```
sudo nmap -O --osscan-guess -T3 -p 22,80 10.10.10.4 -oN 04-os-detection.txt
```

```
sudo nmap -sV -T3 -p 80 --script http-title,http-headers,http-methods 10.10.10.4 -oN  
05-http-enumeration.txt
```

```
ssh-audit --no-colors 10.10.10.4 | tee 06-ssh-audit.txt
```

SAFETY BOUNDARY All probes targeted one owned VM on an isolated network. No brute force, exploitation, denial-of-service testing, or public-system scanning occurred.

6. Evidence: Host Discovery and Port Exposure

```

~ : bash — Konsole
File Edit View Bookmarks Plugins Settings Help
rtt min/avg/max/mdev = 0.776/1.374/2.264/0.568 ms
[user@parrot]~$ nmap --version
Nmap version 7.95 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.7 openssl-3.5.6 libssh2-1.11.1 libz-1.3.1 libpcap-1.10.5 nmap-libdnet-1.12 ipv6
Compiled without:
Available nsock engines: epoll poll select
[user@parrot]~$ mkdir -p ~/SOC-Labs/Nmap-Enumeration
[user@parrot]~$ sudo nmap -sn 10.10.10.4 -oN ~/SOC-Labs/Nmap-Enumeration/01-host-discovery.txt
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-22 18:24 UTC
Nmap scan report for 10.10.10.4
Host is up (0.00077s latency).
MAC Address: 08:00:27:91:C9:2A (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.06 seconds
[user@parrot]~$ cat ~/SOC-Labs/Nmap-Enumeration/01-host-discovery.txt
# Nmap 7.95 scan initiated Sat Aug 22 18:24:52 2026 as: nmap -sn -oN /home/user/SOC-Labs/Nmap-Enumeration/01-host-discovery.txt 10.10.10.4
Nmap scan report for 10.10.10.4
Host is up (0.00077s latency).
MAC Address: 08:00:27:91:C9:2A (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
# Nmap done at Sat Aug 22 18:24:52 2026 -- 1 IP address (1 host up) scanned in 0.06 seconds
[user@parrot]~$

```

Figure 1. Nmap host discovery confirmed 10.10.10.4 was online.

```

# Nmap done at Sat Aug 22 18:24:52 2026 -- 1 IP address (1 host up) scanned in 0.06 seconds
[user@parrot]~$ sudo nmap -sS -T3 --top-ports 100 --reason 10.10.10.4 -oN ~/SOC-Labs/Nmap-Enumeration/02-top-ports.txt
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-22 18:28 UTC
Nmap scan report for 10.10.10.4
Host is up, received arp-response (0.00080s latency).
Not shown: 98 filtered tcp ports (no-response)
PORT      STATE SERVICE REASON
22/tcp    open  ssh      syn-ack ttl 64
80/tcp    open  http     syn-ack ttl 64
MAC Address: 08:00:27:91:C9:2A (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.94 seconds
[user@parrot]~$ cat ~/SOC-Labs/Nmap-Enumeration/02-top-ports.txt
# Nmap 7.95 scan initiated Sat Aug 22 18:28:27 2026 as: nmap -sS -T3 --top-ports 100 --reason -oN /home/user/SOC-Labs/Nmap-Enumeration/02-top-ports.txt 10.10.10.4
Nmap scan report for 10.10.10.4
Host is up, received arp-response (0.00080s latency).
Not shown: 98 filtered tcp ports (no-response)
PORT      STATE SERVICE REASON
22/tcp    open  ssh      syn-ack ttl 64
80/tcp    open  http     syn-ack ttl 64
MAC Address: 08:00:27:91:C9:2A (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

# Nmap done at Sat Aug 22 18:28:29 2026 -- 1 IP address (1 host up) scanned in 1.94 seconds
[user@parrot]~$

```

Figure 2. Controlled top-100 SYN scan found SSH and HTTP open; 98 ports were filtered.

7. Evidence: Service and OS Identification

```

~ : bash — Konsole
File Edit View Bookmarks Plugins Settings Help
Not shown: 98 filtered tcp ports (no-response)
PORT      STATE SERVICE REASON
22/tcp    open  ssh      syn-ack ttl 64
80/tcp    open  http     syn-ack ttl 64
MAC Address: 08:00:27:91:C9:2A (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

# Nmap done at Sat Aug 22 18:28:29 2026 -- 1 IP address (1 host up) scanned in 1.94 seconds
[user@parrot]--[~]
$ sudo nmap -sV -sC -T3 -p 22,80 10.10.10.4 -oN ~/SOC-Labs/Nmap-Enumeration/03-service-versions.txt
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-22 18:32 UTC
Nmap scan report for 10.10.10.4
Host is up (0.00090s latency).

PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 9.6p1 Ubuntu 3ubuntu13.18 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|_ 256 b2:cf:cf:8c:f1:4e:03:4a:01:98:bd:e3:d5:e4:a7:d1 (ECDSA)
|_ 256 b1:e4:60:e2:55:45:68:98:04:de:5b:43:0b:db:d6:64 (ED25519)
80/tcp    open  http     Apache httpd 2.4.58 ((Ubuntu))
|_ http-title: Apache2 Ubuntu Default Page: It works
|_ http-server-header: Apache/2.4.58 (Ubuntu)
MAC Address: 08:00:27:91:C9:2A (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.60 seconds
[user@parrot]--[~]
$

```

Figure 3. Nmap identified OpenSSH 9.6p1 and Apache 2.4.58.

```

0.32 (91%), Linux 2.6.32 - 3.13 (91%), Linux 3.10 - 4.11 (91%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 3.91 seconds
[user@parrot]--[~]
$ cat ~/SOC-Labs/Nmap-Enumeration/04-os-detection.txt
# Nmap 7.95 scan initiated Sat Aug 22 18:36:33 2026 as: nmap -O --osscan-guess -T3 -p 22,80 -oN /home/user/SOC-Labs/Nmap-Enumeration/04-os-detection.txt 10.10.10.4
Nmap scan report for 10.10.10.4
Host is up (0.00100s latency).

PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
MAC Address: 08:00:27:91:C9:2A (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: Linux 4.15 - 5.19 (97%), Linux 4.19 (97%), Linux 5.0 - 5.14 (97%), OpenWrt 21.02 (Linux 5.4) (97%), MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3) (97%), Linux 6.0 (95%), Linux 5.4 - 5.10 (91%), Linux 2.6.32 (91%), Linux 2.6.32 - 3.13 (91%), Linux 3.10 - 4.11 (91%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
# Nmap done at Sat Aug 22 18:36:37 2026 -- 1 IP address (1 host up) scanned in 3.91 seconds
[user@parrot]--[~]
$

```

Figure 4. OS fingerprinting inferred Linux but reported non-ideal test conditions.

Evidence	Observed value	Confidence
OpenSSH	9.6p1 Ubuntu-3ubuntu13.18	Confirmed by endpoint
Apache	2.4.58 (Ubuntu)	Confirmed by endpoint
OS family	Linux	Confirmed
Exact OS/kernel from Nmap	No exact match	Limited by filtered ports

8. Evidence: HTTP Enumeration

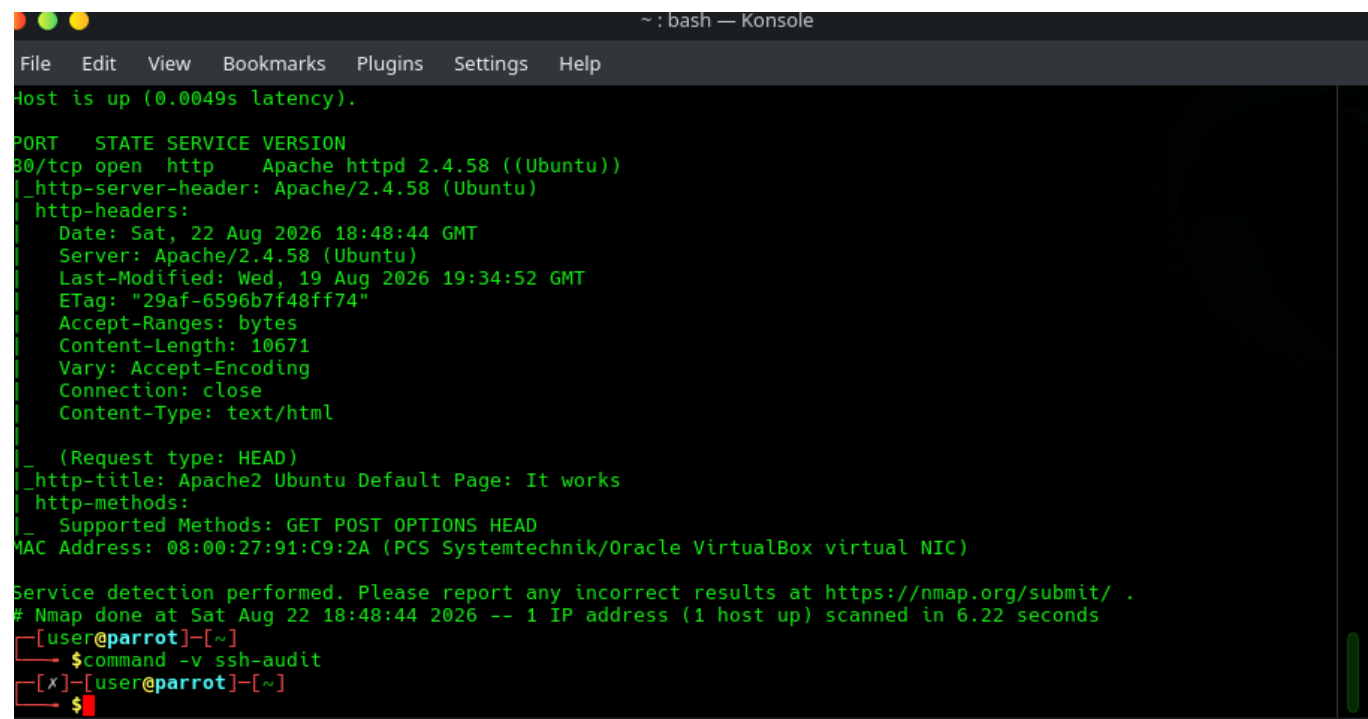


Figure 5. Apache banner, default page, headers, and HTTP methods.

HTTP item	Observed result	Assessment
Server header	Apache/2.4.58 (Ubuntu)	Low-severity information disclosure
Page title	Apache2 Ubuntu Default Page: It works	Default-content hardening opportunity
Methods	GET, POST, OPTIONS, HEAD	Review against application requirements
Security conclusion	No exploit attempt or proof of vulnerability	Informational / hardening

9. Evidence: SSH Cryptographic Assessment

```

~: bash — Konsole
File Edit View Bookmarks Plugins Settings Help
$ ssh-audit -n 10.10.10.4 | tee ~/SOC-Labs/Map-Enumeration/06-ssh-audit.txt
# general
(gen) banner: SSH-2.0-OpenSSH_9.6p1 Ubuntu-3ubuntu13.18
(gen) software: OpenSSH 9.6p1
(gen) compatibility: OpenSSH 9.6+, Dropbear SSH 2020.79+
(gen) compression: enabled (zlib@openssh.com)

# key exchange algorithms
(kex) sntrup761x25519-sha512@openssh.com -- [info] available since OpenSSH 8.5
-- [info] default key exchange from OpenSSH 9.0 to 9.8
-- [info] hybrid key exchange based on post-quantum resistant algo
rithm and proven conventional X25519 algorithm
(kex) curve25519-sha256 -- [info] available since OpenSSH 7.4, Dropbear SSH 2018.76
-- [info] default key exchange from OpenSSH 7.4 to 8.9
(kex) curve25519-sha256@libssh.org -- [info] available since OpenSSH 6.4, Dropbear SSH 2013.62
-- [info] default key exchange from OpenSSH 6.5 to 7.3
(kex) ecdh-sha2-nistp256 -- [fail] using elliptic curves that are suspected as being backdo
ored by the U.S. National Security Agency
-- [info] available since OpenSSH 5.7, Dropbear SSH 2013.62
-- [fail] using elliptic curves that are suspected as being backdo
(kex) ecdh-sha2-nistp384 -- [info] available since OpenSSH 5.7, Dropbear SSH 2013.62
-- [fail] using elliptic curves that are suspected as being backdo
(kex) ecdh-sha2-nistp521 -- [info] available since OpenSSH 5.7, Dropbear SSH 2013.62
-- [fail] using elliptic curves that are suspected as being backdo
(kex) diffie-hellman-group-exchange-sha256 (3072-bit) -- [info] available since OpenSSH 4.4
-- [info] OpenSSH's GEX fallback mechanism was triggered
during testing. Very old SSH clients will still be able to create connections using a 2048-bit modulus, though
modern clients will use 3072. This can only be disabled by recompiling the code (see https://github.com/opens
ssh/openssh-portable/blob/v_9.4/dh.c#L477).
(kex) diffie-hellman-group16-sha512 -- [info] available since OpenSSH 7.3, Dropbear SSH 2016.73
(kex) diffie-hellman-group18-sha512 -- [info] available since OpenSSH 7.3
(kex) diffie-hellman-group14-sha256 -- [warn] 2048-bit modulus only provides 112-bits of symmetric str
ength
(kex) ext-info-s -- [info] available since OpenSSH 7.3, Dropbear SSH 2016.73
-- [info] pseudo-algorithm that denotes the peer supports RFC8308
extensions
(kex) kex-strict-s-v00@openssh.com -- [info] pseudo-algorithm that denotes the peer supports a strict
er key exchange method as a counter-measure
to the Terrapin attack (CVE-2023-48795)

# host-key algorithms
(key) rsa-sha2-512 (3072-bit) -- [info] available since OpenSSH 7.2
(key) rsa-sha2-256 (3072-bit) -- [info] available since OpenSSH 7.2, Dropbear SSH 2020.79
(key) ecdsa-sha2-nistp256 -- [fail] using elliptic curves that are suspected as being backdo
ored by the U.S. National Security Agency
-- [warn] using weak random number generator could reveal the key
(key) ssh-ed25519 -- [info] available since OpenSSH 5.7, Dropbear SSH 2013.62
-- [info] available since OpenSSH 6.5, Dropbear SSH 2020.79

# encryption algorithms (ciphers)
(enc) chacha20-poly1305@openssh.com -- [info] available since OpenSSH 6.5, Dropbear SSH 2020.79
-- [info] default cipher since OpenSSH 6.9
(enc) aes128-ctr -- [info] available since OpenSSH 3.7, Dropbear SSH 0.52
  
```

Figure 6. SSH-Audit identified modern KEX and host-key options plus policy-dependent warnings.

```

(key) ssh-ed25519 -- [info] available since OpenSSH 6.5, Dropbear SSH 2020.79

# encryption algorithms (ciphers)
(enc) chacha20-poly1305@openssh.com -- [info] available since OpenSSH 6.5, Dropbear SSH 2020.79
    ^- [info] default cipher since OpenSSH 6.9
(enc) aes128-ctr -- [info] available since OpenSSH 3.7, Dropbear SSH 0.52
(enc) aes192-ctr -- [info] available since OpenSSH 3.7
(enc) aes256-ctr -- [info] available since OpenSSH 3.7, Dropbear SSH 0.52
(enc) aes128-gcm@openssh.com -- [info] available since OpenSSH 6.2
(enc) aes256-gcm@openssh.com -- [info] available since OpenSSH 6.2

# message authentication code algorithms
(mac) umac-64-etm@openssh.com -- [warn] using small 64-bit tag size
    ^- [info] available since OpenSSH 6.2
(mac) umac-128-etm@openssh.com -- [info] available since OpenSSH 6.2
(mac) hmac-sha2-256-etm@openssh.com -- [info] available since OpenSSH 6.2
(mac) hmac-sha2-512-etm@openssh.com -- [info] available since OpenSSH 6.2
(mac) hmac-sha1-etm@openssh.com -- [fail] using broken SHA-1 hash algorithm
    ^- [info] available since OpenSSH 6.2
(mac) umac-64@openssh.com -- [warn] using encrypt-and-MAC mode
    ^- [warn] using small 64-bit tag size
    ^- [info] available since OpenSSH 4.7
(mac) umac-128@openssh.com -- [warn] using encrypt-and-MAC mode
    ^- [info] available since OpenSSH 6.2
(mac) hmac-sha2-256 -- [warn] using encrypt-and-MAC mode
    ^- [info] available since OpenSSH 5.9, Dropbear SSH 2013.56
(mac) hmac-sha2-512 -- [warn] using encrypt-and-MAC mode
    ^- [info] available since OpenSSH 5.9, Dropbear SSH 2013.56
(mac) hmac-sha1 -- [fail] using broken SHA-1 hash algorithm
    ^- [warn] using encrypt-and-MAC mode
    ^- [info] available since OpenSSH 2.1.0, Dropbear SSH 0.28

# fingerprints
(fin) ssh-ed25519: SHA256:IkdrUpcUZns9ZN3DqZiArUjNHlu4oZLSThH+mEn4pk
(fin) ssh-rsa: SHA256:00LBiPlRPs7m+dkdZ/mGoSNpvryrBF/Xrb6b0BGltD4

# algorithm recommendations (for OpenSSH 9.6)
(rec) -diffie-hellman-group14-sha256 -- kex algorithm to remove
(rec) -ecdh-sha2-nistp256 -- kex algorithm to remove
(rec) -ecdh-sha2-nistp384 -- kex algorithm to remove
(rec) -ecdh-sha2-nistp521 -- kex algorithm to remove
(rec) -ecdsa-sha2-nistp256 -- key algorithm to remove
(rec) -hmac-sha1 -- mac algorithm to remove
(rec) -hmac-sha1-etm@openssh.com -- mac algorithm to remove
(rec) -hmac-sha2-256 -- mac algorithm to remove
(rec) -hmac-sha2-512 -- mac algorithm to remove
(rec) -umac-128@openssh.com -- mac algorithm to remove
(rec) -umac-64-etm@openssh.com -- mac algorithm to remove
(rec) -umac-64@openssh.com -- mac algorithm to remove

# additional info
(info) For hardening guides on common OSes, please see: <https://www.ssh-audit.com/hardening_guides.html>
(info) Be aware that, while this target properly supports the strict key exchange method (via the kex-strict-?-v00@openssh.com marker) needed to protect against the Terrapin vulnerability (CVE-2023-48795), all peers must

```

Figure 7. Cipher, MAC, fingerprint, and removal recommendations.

Category	Observation	Classification
KEX	sntrup761x25519 and Curve25519 available	Strong modern options
Terrapin	Strict KEX extension supported	Mitigation present
Ciphers	ChaCha20-Poly1305, AES-GCM, AES-CTR	Strong / acceptable
MACs	HMAC-SHA1 and 64-bit UMAC remain	Legacy compatibility; remove if not needed
NIST curves	SSH-Audit flags them strongly	Tool-policy warning; not proof of a backdoor

10. Source Validation

```
socstudent@linux-webserver01:~$ hostname
linux-webserver01
socstudent@linux-webserver01:~$ ip -br address
lo                UNKNOWN      127.0.0.1/8 ::1/128
enp0s3            UP           10.10.10.4/24 metric 100 fe80::a00:27ff:fe91:c92a/64
socstudent@linux-webserver01:~$ sudo systemctl is-active ssh
[sudo] password for socstudent:
Sorry, try again.
[sudo] password for socstudent:
inactive
socstudent@linux-webserver01:~$ sudo systemctl is-active apache2
active
socstudent@linux-webserver01:~$ sudo systemctl enable --now ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink /etc/systemd/system/ssh.service + /usr/lib/systemd/system/ssh.service.
Created symlink /etc/systemd/system/multi-user.target.wants/ssh.service + /usr/lib/systemd/system/ssh.service.
socstudent@linux-webserver01:~$ sudo systemctl is-active ssh
active
socstudent@linux-webserver01:~$ sudo systemctl enable --now ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
socstudent@linux-webserver01:~$ sudo systemctl is-active ssh
active
socstudent@linux-webserver01:~$ sudo nmap -sn 10.10.10.4 -oN ~/SOC-Labs/Nmap-Enumeration/01-host-discovery.txt
sudo: nmap: command not found
socstudent@linux-webserver01:~$ hostname
linux-webserver01
socstudent@linux-webserver01:~$ sudo ss -ltnp | grep -E ':(22|80)[[:space:]]'
[sudo] password for socstudent:
LISTEN 0      4096      0.0.0.0:*        users:((("sshd",pid=2097,fd=3),("systemd",pid=1,fd=144))
LISTEN 0      511      *:80             users:((("apache2",pid=899,fd=4),("apache2",pid=898,fd=4),("apache2",pid=896,fd=4))
LISTEN 0      4096      [::]:22         [::]:*        users:((("sshd",pid=2097,fd=4),("systemd",pid=1,fd=146))
socstudent@linux-webserver01:~$ ssh -V
OpenSSH_9.6p1 Ubuntu-3ubuntu13.18, OpenSSL 3.0.13 30 Jan 2024
socstudent@linux-webserver01:~$ apache2 -v
Server version: Apache/2.4.58 (Ubuntu)
Server built:   2026-07-06T16:27:55
socstudent@linux-webserver01:~$ sudo sshd -T | grep -E '^(\cipher|macs|kexalgorithms)'
ciphers chacha20-poly1305@openssh.com,aes128-ctr,aes192-ctr,aes256-ctr,aes128-gcm@openssh.com,aes256-gcm@openssh.com,
macs umac-64-etm@openssh.com,umac-128-etm@openssh.com,hmac-sha2-256-etm@openssh.com,hmac-sha2-512-etm@openssh.com,hmac-sha1-etm@openssh.com,umac-64@openssh.com,
umac-128@openssh.com,hmac-sha2-256,hmac-sha2-512,hmac-sha1
kexalgorithms sntrup761x25519-sha512@openssh.com,curve25519-sha256,curve25519-sha256@libssh.org,ecdh-sha2-nistp256,ecdh-sha2-nistp384,ecdh-sha2-nistp521,diffie-
hellman-group-exchange-sha256,diffie-hellman-group16-sha512,diffie-hellman-group18-sha512,diffie-hellman-group14-sha256
socstudent@linux-webserver01:~$
```

Figure 8. Ubuntu source evidence validated listening services, versions, and effective SSH algorithms.

Validation question	Authoritative result
What OS was actually running?	Ubuntu 24.04.4 LTS; kernel 6.8.0-138-generic
What owned ports 22 and 80?	sshd and apache2 on IPv4 and IPv6 listeners
Did version detection match?	Yes - OpenSSH 9.6p1 and Apache 2.4.58
Did SSH-Audit match effective config?	Yes - sshd -T listed the observed KEX, ciphers, and MACs

ANALYST JUDGMENT Automated scanner output was treated as evidence to validate, not unquestioned truth. Nmap's OS limitation and SSH-Audit's policy opinions were explicitly preserved in the final assessment.

11. Evidence Integrity

```

[user@parrot]~$ cd ~/SOC-Labs/Nmap-Enumeration
[user@parrot]~/SOC-Labs/Nmap-Enumeration$ sha256sum 01-host-discovery.txt 02-top-ports.txt 03-services.txt 04-os-detection.txt 05-http-enumeration.txt 06-ssh-audit.txt | tee SHA256SUMS.txt
sha256sum: 9cda4b9adb442a6b1ef94514cc73c9f93f53e7851635a97ddc8692298566dae1 01-host-discovery.txt
2a32a4025dd6f3341396ee33bb84922ca9cb8de5c7c6fe5f271340483ed3bd98 02-top-ports.txt
03-services.txt: No such file or directory
cdda307418c74f28794dcd779f367666d0f45ad79b413881ea86e1016921085a 04-os-detection.txt
c9e3922ea4a40e05a4113734eb8f9f861f3031cd2e72c870fa15e2130ef88e9 05-http-enumeration.txt
2513a08c288a8c48865c5bd44c19ef5469ccbc30196ad09b5aab5d857693fb66 06-ssh-audit.txt
[user@parrot]~/SOC-Labs/Nmap-Enumeration$

```

Figure 9. SHA-256 hashes generated for the six assessment evidence files.

Hashing establishes a reproducible integrity check. A later reviewer can calculate SHA-256 again and compare the values to determine whether an evidence file changed after collection.

12. Findings and Risk Classification

Finding	Severity	Rationale
SSH and HTTP exposed	Expected	Required lab services; restrict by business and network need
Precise service banners	Low	Assists attacker version research but does not itself establish exploitability
Default Apache page	Informational	Reveals default deployment and unnecessary content
Legacy SSH MAC/KEX support	Low-Medium	Modern alternatives exist; remove legacy options where compatibility permits
OS fingerprint limitation	Defensive benefit	Filtering reduced Nmap's ability to produce an exact match
Evidence of compromise	None observed	Assessment enumerated exposure only; no exploitation performed

13. Recommended Remediation

- Keep only services that have a documented requirement; restrict SSH source networks with UFW or an upstream control.
- Prefer public-key authentication and disable password authentication when operationally appropriate.
- Remove HMAC-SHA1, 64-bit UMAC, and unnecessary non-Encrypt-then-MAC algorithms after compatibility testing.
- Remove the Apache default page and reduce unnecessary server-version disclosure.
- Review whether POST and OPTIONS are needed for the deployed application.
- Retest after changes and compare the new results with this hashed baseline.

14. MITRE ATT&CK; Context

MITRE ATT&CK; T1046, Network Service Discovery, is relevant to the observable behavior because Nmap probed a host to identify available network services. In this lab the activity was authorized, constrained to one owned VM, and produced no exploitation. A defender should correlate similar scanning with asset ownership, source authorization, breadth, speed, and any subsequent authentication or exploitation activity before assigning malicious intent.

15. Skills Demonstrated

- Authorized scope definition and scan safety
- Nmap host discovery, SYN scanning, version detection, NSE, and OS fingerprinting
- SSH-Audit cryptographic configuration assessment
- HTTP banner and method analysis
- Firewall-behavior interpretation
- Scanner-to-endpoint source validation
- Risk classification and evidence-based reporting
- SHA-256 evidence integrity preservation

16. Employer-Facing Project Summary

RESUME BULLET Performed an authorized Nmap and SSH-Audit assessment of an isolated Ubuntu server; identified SSH/HTTP exposure, validated service versions and cryptographic configuration against host evidence, classified hardening opportunities, and preserved six evidence files with SHA-256 hashes.

Interview talking points

- Why filtered ports reduced OS-fingerprint confidence.
- How I separated a tool warning from a confirmed vulnerability.
- Why source validation made the final findings more defensible.
- How I preserved evidence integrity and constrained assessment scope.

17. Assistance and Source Transparency

This project was completed by Ryan Bryant in his personal home lab with assistance from OpenAI ChatGPT (Sarah). ChatGPT helped plan the safe workflow, troubleshoot command and VirtualBox issues, explain scan output, locate official documentation, and edit this portfolio narrative. Ryan personally configured and operated the VMs, entered commands, ran the tools, generated the evidence, validated the findings, captured screenshots, and reviewed the conclusions. ChatGPT did not independently access or operate the lab computer.

18. References and Validation Sources

[Nmap Network Scanning - Host Discovery](#). Official explanation of -sn and discovery behavior.

[Nmap Network Scanning - Service and Version Detection](#). Official explanation of service probes and version matching.

[Nmap Network Scanning - OS Detection](#). Official documentation noting improved accuracy with at least one open and one closed TCP port.

[SSH-Audit maintained project documentation](#). Tool capabilities, algorithm analysis, and recommendation behavior.

[Ubuntu Server - OpenSSH server](#). Official service, configuration, validation, and banner-disclosure guidance.

[Ubuntu Server - OpenSSH crypto configuration](#). Official cryptographic algorithm configuration guidance.

[MITRE ATT&CK; T1046 - Network Service Discovery](#). Technique context for service-discovery activity.

FINAL DISPOSITION Authorized security assessment completed. Two expected services were confirmed; several low-to-medium hardening opportunities were documented; no exploitation or evidence of compromise was observed.